KEIM Australia Master Application Manual

This master manual brings together the site application standards for **KEIM Innoprim, KEIM Innostar, KEIM Soldalit, KEIM Concretal-Lasur and KEIM Twinstar**. To validate the KEIM Australia project warranty, all works must follow the product sections below and the current Technical Data Sheet (TDS) for each product. Separate, more detailed guides are available for Soldalit, Innostar and Concretal-Lasur at keim.com.au.

Part 1: Technical Reference Tables

1. Airless Spray Set-up

Use a new or dedicated spray hose. The high alkalinity of sol-silicate paints loosens old, dried acrylic skins from the inside of used lines, causing repeated tip blockages.

Product	Tip (minimum)	Gun filter	Setting	Pump	Follow-up
KEIM Innoprim	≥ 423	30 mesh	> 90 bar (gun closed)	Piston or diaphragm	Back-roll immediately on bare substrates
KEIM Innostar	≥ 421	30 mesh	Heat material to 40°C (in-line heater); set pressure for an even, tail-free fan	Piston or diaphragm	Back-roll immediately on bare substrates
KEIM Soldalit	≥ 435	30 mesh	> 85 bar	**Piston only**	Back-roll the first coat immediately
KEIM Soldalit-Grob	≥ 443	No filter	> 110 bar	**Piston only**	Back-roll the first coat immediately
KEIM Concretal-Lasur	≥ 512	30 mesh	> 170 bar	Piston or diaphragm	Lay off immediately with a brush or unloaded roller
KEIM Twinstar	≥ 539	No filter	> 90 bar	Piston or diaphragm	Immediate vertical roller pass

Pump selection: KEIM Soldalit and KEIM Soldalit-Grob must be sprayed with a piston pump. Do not use diaphragm pumps for these two products.

2. Roller & Stroke Technique

Product	Roller	Stroke pattern	Edges & joins
KEIM Innostar	12–15mm short to medium-nap polyamide. No foam rollers.	True cross-hatch in all directions, finished with a light vertical lay-off pass.	Keep a continuous wet edge. No spot-priming or touch-ups over dry areas (causes permanent halos).
KEIM Soldalit	15–20mm long-pile polyamide. 20mm suits very rough render; use 15mm on finer finishes to avoid drag marks.	Firm, multi-directional cross-hatch to avoid tracking and roller banding.	Keep a continuous wet edge. Work down scaffold lifts to natural architectural breaks. No spot-priming.
KEIM Concretal-Lasur	Short to medium-nap roller, or a quality glaze brush (preferred for touch-ups).	Random criss-cross to spread the pigment evenly, then lay off lightly vertically.	Work from the bottom upward on walls so runs merge cleanly. Never cut in.
KEIM Twinstar	Quality long-pile polyamide.	Wet-on-wet: cross-hatch first, then immediately roll a second wet layer vertically.	Complete each section in one go with no drying in between. Keep a wet edge across whole sections.

Part 2: Product Application Profiles

Profile 1: KEIM Innostar & KEIM Innoprim (Interior)

Coat configurations. KEIM Innostar is a thin-film mineral coating and does not fill or build out surface texture.

- **New / bare plasterboard:** Apply ONE coat of KEIM Innoprim followed by ONE coat of KEIM Innostar. A second topcoat is often needed in critical or glancing light to build a uniform roller stipple over jointing compound.
- **Deep / dark colours (Price Groups 2 & 3):** A third system coat may be required for full opacity and an even, shadow-free finish in critical light.
- **Coat count reset:** If other trades plaster, patch or sand after the first KEIM coat, the coat count for that area starts again. Re-apply the full system corner to corner over the whole wall section.
- **Even-absorbency substrates:** ONE to TWO coats of KEIM Innostar, depending on hiding.
- **Sound previously painted walls:** ONE coat of KEIM Innostar is typically enough.

Preparation.

- **Plaster finish (AS/NZS 2589):** Walls and ceilings in glancing or critical light need a Level 5 finish. Over Level 3 or 4 finishes in critical light, the thin film will show texture differences, and opacity or banding claims will not be accepted.
- **General:** Substrates must be sound, dry and free of grease, release oils and loose material.
- **Repairs:** Sand patches smooth with 120–180 grit. This covers standard gypsum, vinyl and polymer fillers.
- **Dust removal (critical):** Vacuum, then wipe the whole surface with a clean, slightly damp microfibre cloth. Sol-silicate paint will not bond to sanding dust.
- **Gloss Surfaces:** Any gloss, satin, or shiny surfaces should be mechanically sanded back to a dull finish using sandpaper or an abrasive pad to create a sound physical key. Where a sound, intact acrylic or organic surface cannot be fully sanded, a cross-hatch adhesion trial is required.
  Cross-Hatch Adhesion Trial (painting over acrylic or organic surfaces): (1) Select a representative 300mm × 300mm test area on the existing surface. (2) Clean thoroughly and apply KEIM at the standard rate without dilution. (3) Allow to cure for a minimum of 24 hours. (4) Using a sharp blade, score a 6 × 6 grid of cuts (10mm spacing) through the full KEIM film down to the existing surface. (5) Apply a strip of high-tack tape (e.g. 3M 610 or equivalent) firmly over the scored grid, then pull off sharply at 60°. (6) If fewer than 5% of the scored squares show any delamination, adhesion is acceptable and full application may proceed. If significant areas pull away, the existing coating must be mechanically abraded back to a dull, sound key before application.

Conditions & drying.

- **Temperature:** 5°C to 30°C, air and substrate.

- **Moisture:** Surfaces must be wind-dry, with a moisture content below 8% by weight (5% or less is ideal). Do not paint at 8% or above: find and fix the source of moisture first.
- **Between coats:** At least 6 hours.
- **Drying appearance:** Mineral coatings dry unevenly and lighten as they cure. Do not touch up while drying.

Profile 2: KEIM Soldalit (Exterior)

Coat configurations.

- **Minimum two coats:** One base coat and one topcoat. Single-coat applications over bare, unprimed or patched exterior substrates are not covered by the warranty.
- **Topcoat:** Whether the base coat is KEIM Soldalit, Soldalit-Grob or Contact-Plus, the finish coat is always standard KEIM Soldalit.
- **Coverage:** Approx. 0.35–0.5 kg/m² for two coats on smooth surfaces. Rough surfaces use more. Always confirm on a test area.

Preparation.

- **Gloss Surfaces:** Any gloss, satin, or shiny surfaces should be mechanically sanded back to a dull finish using sandpaper or an abrasive pad to create a sound physical key. Where a sound, intact acrylic or organic surface cannot be fully sanded, a cross-hatch adhesion trial is required.
  Cross-Hatch Adhesion Trial (painting over acrylic or organic surfaces): (1) Select a representative 300mm × 300mm test area on the existing surface. (2) Clean thoroughly and apply KEIM at the standard rate without dilution. (3) Allow to cure for a minimum of 24 hours. (4) Using a sharp blade, score a 6 × 6 grid of cuts (10mm spacing) through the full KEIM film down to the existing surface. (5) Apply a strip of high-tack tape (e.g. 3M 610 or equivalent) firmly over the scored grid, then pull off sharply at 60°. (6) If fewer than 5% of the scored squares show any delamination, adhesion is acceptable and full application may proceed. If significant areas pull away, the existing coating must be mechanically abraded back to a dull, sound key before application.
- **Stripping:** Poultice-based strippers are not recommended; they can draw damaging salts to the surface. Use KEIM Dispersion Remover to strip acrylics.
- **Water absorption test (runners test):** Mist water onto the dry wall.
  - **Short, thick tracks (highly absorbent):** Prime with undiluted KEIM Soldalit-Fixativ (approx. 0.15 L/m²).
  - **Long, thin tracks (normal):** Go straight to the first coat.
  - **Beading (non-absorbent):** A water-repellent sealer is present and must be removed mechanically first.
- **New render & repairs:** Check new render and repair areas for a glassy, sintered surface skin. If one is present, contact KEIM Australia on (02) 9748 7160 before painting.
- **Existing coating soundness:** Score a "#" into the old coating, apply high-tack tape and pull off sharply. If it lifts, scrape back to a sound edge before painting.
- **Lead warning:** Take care removing pre-1960s coatings; they may contain lead.

Patching.

- **Texture matching (critical):** The paint copies the texture beneath it. Patches must match the surrounding render exactly or they will show in direct sun.
- **Fillers:** Use mineral (cement-based) fillers and repair mortars. Gypsum, polymer or acrylic fillers risk porosity and expansion mismatch, which can show as flashing or mapping. Before using any filler, do a trial: apply it to a 300mm × 300mm area, cure fully, paint at the standard rate, then check adhesion (tape pull), colour evenness and cracking. Only proceed if the trial passes.

Priming & dilution.

- **Primer:** Apply undiluted KEIM Soldalit-Fixativ, working from the bottom of the wall upward to avoid run marks.
- **Base coat:** May be thinned with up to 5% KEIM Soldalit-Fixativ, or up to 10% on highly absorbent substrates (approx. 0.25–0.5 kg per 5 kg drum). Soldalit-Grob: maximum 5%. The topcoat is applied undiluted.
- **No water:** Never thin KEIM Soldalit with water.

Base coat selection.

- **Sound, even surfaces:** KEIM Soldalit.
- **Hairline cracks up to 0.5mm:** KEIM Soldalit-Grob as the first coat (approx. 0.25 kg/m²).
- **Cracks 0.5–1mm or heavy texture:** KEIM Contact-Plus as the base coat.

Conditions, drying & maintenance.

- **Temperature:** At least 5°C during application and drying (KEIM TDS). KEIM Australia also recommends not applying above 30°C.
- **Moisture:** Surfaces must be wind-dry, with a moisture content below 8% by weight (5% or less is ideal). Do not paint at 8% or above: find and fix the source of moisture first.
- **Sun:** Avoid facades heated directly by the sun; fast drying causes lap marks.
- **Between coats:** At least 12 hours (at 23°C, 50% RH). Do not touch up while drying.
- **Cleaning:** Rinse the facade every 12–24 months with a garden hose or a washer limited to 400 psi. Never use turbo heads.
- **Refresh coat:** After a low-pressure clean, a single refresh coat of standard KEIM Soldalit at approx. 6 m²/kg renews the finish without stripping.

Profile 3: KEIM Concretal-Lasur (Concrete Stain)

KEIM Concretal-Lasur is a thin, penetrating sol-silicate concrete stain for vertical walls and soffits. It is not a floor coating and does not take foot or vehicle traffic. It can be used opaque, to cover blemishes with solid colour, or as a translucent glaze that keeps the concrete's texture and formwork character while hiding repairs.

Preparation.

- **Concrete maturity:** Allow new concrete to cure for at least 28 days. If work must start earlier, confirm moisture is below 4% at 5mm depth first.
- **Release agents:** Remove shuttering oils, curing membranes, grease and laitance completely with a suitable concrete cleaner or degreasing wash, then rinse thoroughly with clean water.
- **Markings:** Grind off or remove all pencil, crayon and spray-paint marks. They will show through a translucent glaze.
- **Glassy skins:** Abrade shiny formwork skins with sand or emery paper to open the surface.
- **Lead warning:** Take care removing old coatings; they may contain lead.

Porosity, moisture & patching.

- **Runners test:**
  - **Short, thick tracks (highly absorbent):** Prime with undiluted KEIM Concretal Fixative (not mixed with pigment) by brush, approx. 0.15 L/m². Do not use tinted glaze to even out absorbency.
  - **Long, thin tracks (normal):** Go straight to the first tinted coat.
  - **Beading:** A water-repellent sealer is present and must be removed mechanically.
- **Moisture:** Concrete must be wind-dry, below 4% at 5mm depth.
- **Patching:** Keep repairs to a minimum. Small pinholes need no filling. Do not feather larger patches beyond the hole. Use open, porous cementitious repair mortars. Film-forming or acrylic fillers seal the surface and leave permanent pale spots. Trial any other compound first and do not use it if the dried stain looks flat or uncoloured over it.

Mixing.

- **Supply formats:** Either a kit with pigment supplied separately from the liquid binder, or pigment only, mixed with KEIM Concretal Fixative as the binder. KEIM Concretal Fixative is the only approved product for mixing and thinning. Transfer all the pigment and rinse the container with a little Fixative.
- **Settling (critical):** The mixed stain is as thin as milk and the mineral pigment settles quickly. Stir thoroughly and often, before and during use.
- **No water:** Never thin with water. Water is for cleaning tools only.

- **Application:** Brush, roller or airless. After rolling or spraying, lay off immediately with a clean brush. Brushing is best for touch-ups.

Conditions & drying.

- **Temperature:** 5°C to 30°C, air and concrete. Do not apply in rain or when rain is expected.
- **Sun:** Work in shade; hot concrete shortens open time and causes lap marks.
- **Between coats:** At least 12 hours (at 23°C, 50% RH).
- **Drying appearance:** The stain looks patchy while drying. Do not touch up during drying.
- **Protection:** Cover glass, metal and glazed ceramics. Wash off splashes immediately with clean water.

Profile 4: KEIM Twinstar (2-in-1 Facade Coating)

KEIM Twinstar is a sol-silicate facade coating whose primer and topcoat are applied in one step, with no drying time in between. It is biocide-free, meets DIN 18363 2.4.1 for dispersion silicate paints, and suits mineral, organic and mixed substrates, including intact ETICS. It is available in light and medium shades.

- **Preparation:** The substrate must be dry, sound, clean and free of chalking. Remove loose coatings. Fill and texture-match defects with a suitable repair material. For new render or render repairs, contact KEIM Australia on (02) 9748 7160 before painting.
- **Dormant cracks up to 0.25mm:** Coat over as normal.
- **Dormant cracks 0.25–0.5mm:** Apply an extra coat of KEIM Twinstar along the crack before the main coating.
- **Mixing:** Stir thoroughly. Always apply undiluted.
- **Application (minimum 0.4 kg/m²; typical consumption approx. 0.45 kg/m²):** (1) Spread the paint criss-cross with a long-pile roller. (2) Immediately roll more material into the wet layer, vertically in one direction.
- **Spraying:** Tip ≥ 539, with the gun filter removed.
- **Conditions:** At least 5°C during application and drying. Do not apply in direct sun or onto sun-heated walls. Protect from sun, wind and rain during and after application.
- **Unsuitable substrates:** Salt-efflorescing surfaces, weathered horizontal or sloping surfaces, timber, elastomeric coatings, saponifiable or water-repellent substrates.

Part 3: Quality Assurance & Warranty Checklist

To validate the KEIM Australia project warranty, the site supervisor should keep the following records:

1. **Proof of purchase:** Tax invoices from KEIM Australia matching the project quantities.
2. **Batch records:** Photographs of all batch numbers on lids and pigment containers.
3. **Site application log:** Daily temperatures, moisture readings, cleaning steps, trial dilutions and confirmation of a dedicated spray hose. Compare actual usage with the quantities from the KEIM Australia calculators.
4. **TDS compliance declaration:** A signed statement that preparation, drying times, mixing and application followed this manual and the current Technical Data Sheets.

Calculators & guides: keim.com.au/calculators. For anything not covered here, contact KEIM Australia on (02) 9748 7160.
